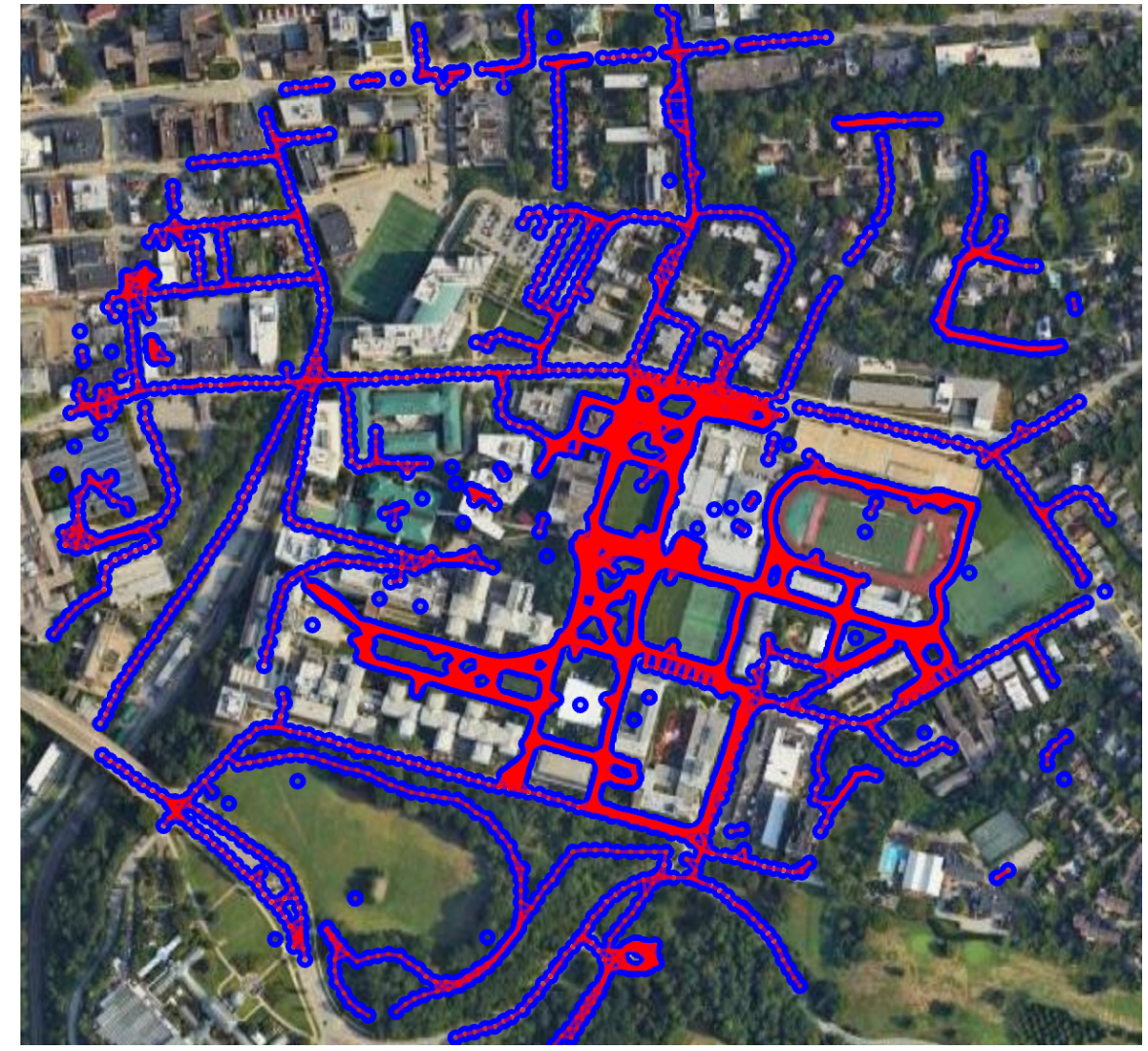
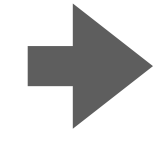
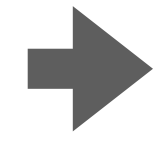
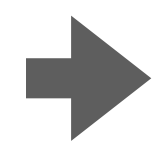
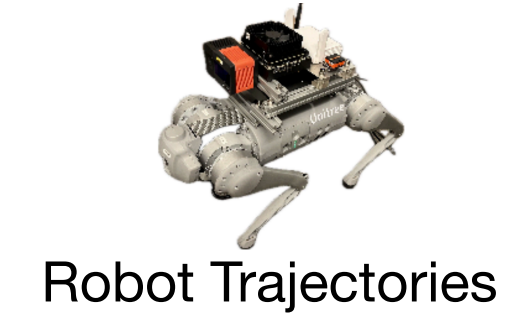


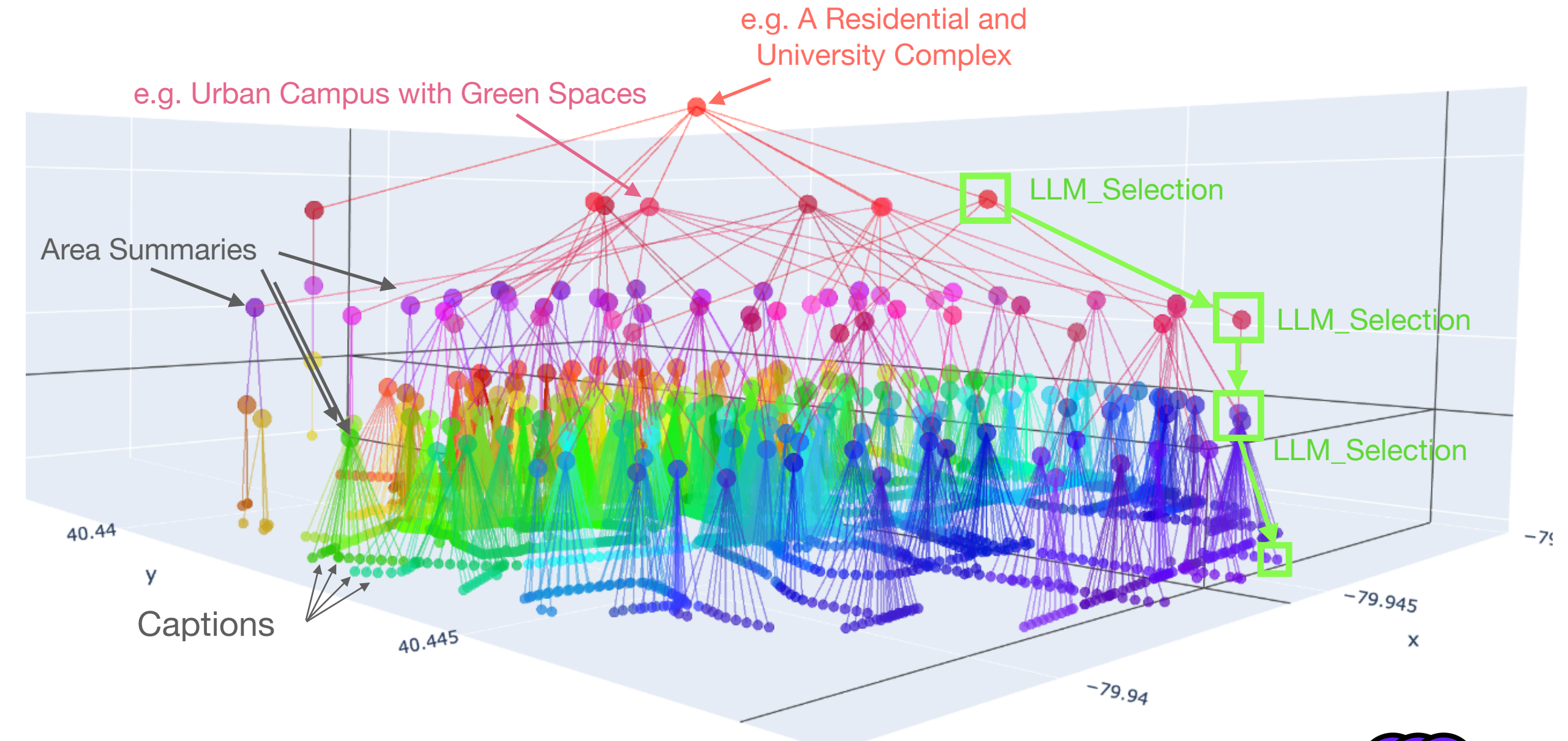
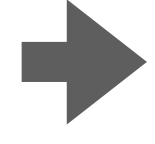
$$E_t = (\tau_t, \mathbf{s}_t, \mathbf{p}_t)$$



Multimodal Graph Node

Connected Edges

Shared Embodied Experiences



Bottom-up Memory Building

Semantic Forest Memory

Retrieval

Top-K Retrieved Results

LLM

Generate

“The closest stairs is at (x,y,z)” → *or* *“The functionality of the building is a multi-purpose ...”*

Environment Aware Answers

1

Global



Can you describe the function of the building?

2

Implicit Queries



Where can I buy drinks?

3

Explicit Queries



Find me the stairs

Natural Language Interface